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Solid Angle (spacial direction)

Solid Angle (rotational measurement of spherical direction): [1]

$$\Omega = \frac{A}{R^2} = \int\limits_A \frac{\cos(\varepsilon)}{R^2}\, dA \qquad \in [0, 4\pi] \tag{1}$$

where $A \equiv$ spherical Area, $R \equiv$ spherical Radius (radial distance), and $\varepsilon = d\vec{A} \cdot \vec{e}_n = \angle(dA, R)$ inclination angle of dA to R along the curved surface A

$$dA = \frac{R^2}{\cos(\varepsilon)} \sin(\theta) d\theta d\varphi = \frac{R^2}{\cos(\varepsilon)} d\Omega \tag{2}$$

where $\varphi \equiv$ horizontal angle, $\theta \equiv$ vertical angle

Symmetry of wave propagation from electromagnetic radiation in vacuum (Reversibility of Beam according

$$d\Omega_g = \frac{\mathcal{M}}{\mathcal{J}} dA_g = \frac{dA_b}{R} \frac{\cos(\varepsilon_b)}{R} = \frac{dA_b \cos(\varepsilon_b)}{R^2} \tag{3}$$

$$d\Omega_b = \frac{\mathcal{M}}{\mathcal{J}} dA_b = \frac{dA_g}{R} \frac{\cos(\varepsilon_g)}{R} = \frac{dA_g \cos(\varepsilon_g)}{R^2} \tag{4}$$

$$d\Omega = \sin(\theta) d\theta d\varphi = \frac{dA_\perp}{R^2} = \frac{\cos(\varepsilon)}{R^2} dA = \mathcal{J} \cdot d\Omega_g = \mathcal{M} \cdot dA_g = \mathcal{J} \cdot \mathcal{M}_b \cdot \frac{\varepsilon_b}{R^2} \tag{5}$$

where $\mathcal{J} \equiv$ Flux Density, $\mathcal{M} \equiv$ Radiant Exitance

[1] $[\Omega] = [sr]$ Steradiant, "Raumwinkel"

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